# Summer Development Log

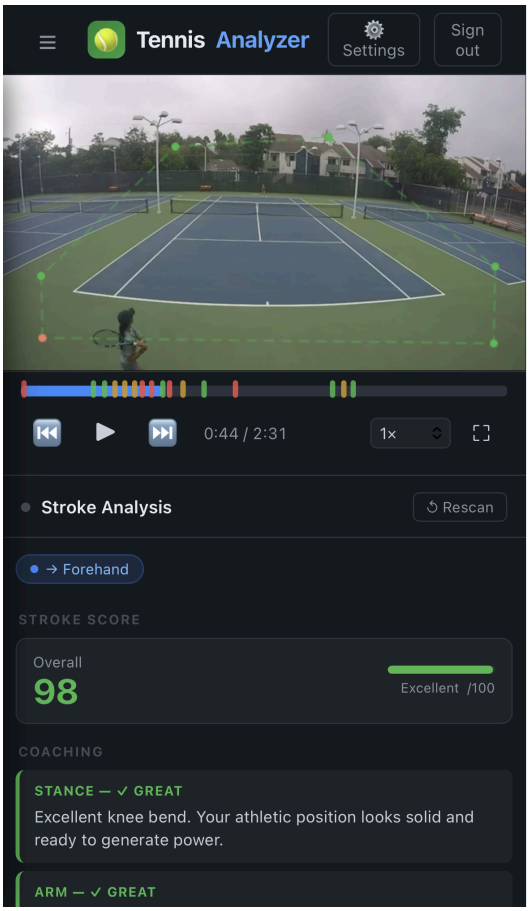
This log documents the majority of work that was done on the Tennis Stroke Analyzer application during the summer. It includes major feature improvements, experiments, debugging, and failed approaches. Testing and finalization continued after this period, with all final code and functionality available in the GitHub repository.

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**— 7/8/26 —**

Achievements

- I made slight changes to the index.html file so that users could also open the page on a mobile device. The same features exist, but the side video library panel becomes collapsible, and the video is displayed above the stroke feedback. The mobile formatting is also adjusted accordingly when the user switches from portrait to landscape orientation.
- Feature improvements:
    - Instead of automatic formatting, the user can choose between the desktop and mobile formats when opening the webpage. The format option is also added to the settings window so the user can switch back and forth without reloading.
    - Hide the angle measurement table in the feedback panel by default (because it takes up too much space in the mobile version). The user can click “show” or “hide” to display the measurements.
    - Added a full-screen button in the corner that enlarges the video.
    - Simplified the controls bar in the mobile version because the buttons didn’t fit. The four skip/rewind buttons were reduced to two, and the ball detected indicator was removed.
    - Sync the play area between the desktop and mobile versions. Previously the outline wasn’t stored, but now the area only needs to be drawn on one device, and it will appear whenever the webpage is opened on another device.



    - Sometimes a body joint measurement will be hidden if the model has low visibility of the relevant landmarks, and the angle won't be shown in the table. Now that the table is able to be hidden this isn't necessary. All angles are shown in the table at all times, but will be greyed out with a small message that says low visibility if that's the case.

Issues

- The overlay that the desktop version used didn't work for mobile because the video screen is much smaller. The body landmark and ball detection dots had to be reduced in size.

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**— 7/4/26 —**

Achievements

- Feature improvements:
    - Whenever the ball detection feature detects many clusters of motion, this usually means that there has been a large disturbance or the camera has shifted. When a frame detects more than 10 new clusters of motion, the frame is skipped and no new circles are displayed. A small alert appears that tells the user ball detection is unreliable.
    - I created a settings button that opens up a window in the middle of the screen with a list of various controls that used to be scattered throughout the page. This includes dominant hand selection, auto-pause after each stroke, on/off switch for ball detection, and play area marking.
    - Switched from MediaPipe complexity 1 to 2 to try to increase the accuracy of the body landmark detection. It resulted in a slightly less smooth overlay, but no noticeable differences.
    - When 3 consecutive frames detect excessive motion, an alert appears that notifies the user the camera may have moved.
    - The user has the option to delete video files that they have uploaded through the side navigation panel. A confirmation window prompts the user to permanently delete or cancel.
    - Cleaned up the stroke detection code so that there must be a minimum of 2 frames that satisfies the wrist speed threshold before a stroke is detected.
    - Added an on/off switch in the settings window that allows the user to choose whether they want angle measurements to appear along with the body landmarks on the video.

Issues

- Whenever a portion of unnecessary code is deleted, random bugs often appear because there are exceptions being thrown that prevent parts of the page from loading. For example, the video library showed that no files were uploaded even though all files are stored, and no colored stroke markers appeared along the progress bar (those are also stored).

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**— 6/29/26 —**

Achievements

- Algorithm modifications:
    - Continued working on ball detection
        - The ball should be much smaller than the player and opponent, and should contain less pixels. I added a maximum pixel count so that motion is ignored when it's too large to be the ball.
        - I removed the cell grid completely. Sometimes clusters would span multiple cells, but if only a few pixels were in a particular cell, it would fail to meet the minimum requirement, and the cell would be ignored, cutting off part of the cluster. All pixels are identified and grouped into clusters before being filtered with the min/max thresholds.
        - Instead of using only a number as the maximum cluster size, I set limitations on the height and width of clusters. The ball should appear on screen as a circular object, so clusters that are very long or wide are clearly not balls and can be ignored. (To check the functionality, I included bounding boxes in the overlay that outline clusters and show the number of pixels in the cluster.)
        - I adjusted the algorithm to determine clusters of pixels. I expanded the distance requirement, so that pixels that were close together, even if they were separated by 1-2 pixels, would still be in the same cluster.
        - I placed a mask surrounding the player's body landmark skeleton so that motion wouldn't be detected in the surrounding area. This prevented the motion pixels from covering up the body landmarks and also reduced the amount of motion that was detected from the player's movement.

Issues

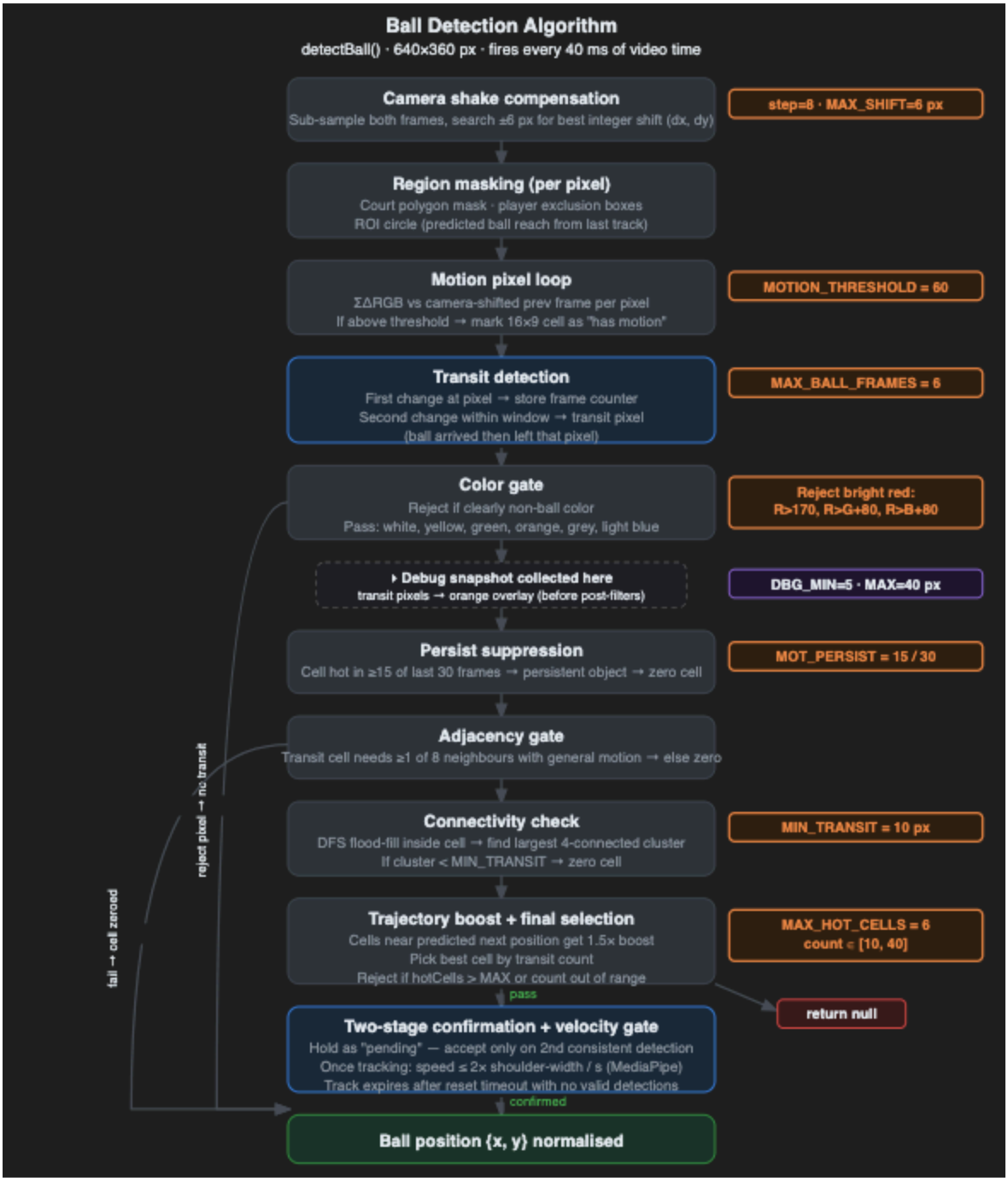
- When the motion detection overlay placed dots near the player's body, it would sometimes override the body landmarks, causing the skeleton to not appear.
- In an attempt to clean up code and simplify the algorithm, I rearranged and deleted code within the ball detection function. However some variables were linked to other parts of the application and the changes affected the cached stroke scan results. All videos had to be rescanned and the accuracy of the ball detection algorithm also fell. I had to retrace my steps and re-implement some of the changes I made to the algorithm earlier to improve accuracy.
- After one of the changes I made, I deleted a section of code which seemed unnecessary, but ended up interfering with the cached scan results. The previously scanned videos were reset, and stroke markers stopped appearing along the progress bar. The problem was resolved after rescanning all the videos.

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**— 6/25/26 —**

## Achievements

- Algorithm modifications:
    - Continued working on ball detection
        - Divide the video screen into individual cells (40x40 pixels) that are scanned for motion. A minimum of 10 pixels within a cell must be triggered with color change for the cell to record motion, and an adjacent cell must have also had motion in the last few frames.



        - To debug, I drew the cell borders over the playback screen. While the video is playing, a cell is highlighted in orange whenever the pixel motion requirement is met. Within the cell, an orange dot appears at the average position of all triggered pixels. When multiple cells light up, the orange dots should show the approximate path of the ball.
        - When the ball is visible, the cells light up accordingly, but the ball is not detected, so the problem likely lies in the color check (initial and final color should be similar). I completely removed the color check, with motion only requiring two consecutive color changes.
        - Random color changes within the video (resulting from poor quality or external movement) causes misdetection. I tried to limit the number of false positives by only selecting the top 2 cells that contain the most triggered pixels, which should relate to clearer motion.
        - I experimented with testing the ball detection at 0.5x speed, and there were better results because the slower speed results in more frames being checked for motion. I adjusted so that all speeds returned the same results.
        - At this point I combined the previous ball detection (detected with motion and color) and the debug trail (incorporates cells and takes average position of triggered pixels). Instead of one trail (initial overlay), multiple possible ball paths are drawn because the debug trail takes the 2 cells with the most motion.
        - To improve accuracy, I attempted a color check again. The check only eliminates pixels that are very clearly not the right color (for example bright red, purple, black, etc.).
- Feature improvements:
    - Every time a video file is added or opened for the first time, the user is given the option to calibrate. Initially I made it so that the user can trace the tennis court, but many times the court will extend beyond the frame. I changed the calibration to a general “play area”. The user can click on the screen to place dots, and after placing multiple, an enclosed polygon is formed. The detection features only check the enclosed area for the player and the ball. The play area is saved so the user doesn’t have to re-calibrate unless they choose to do so.

Issues

- Sometimes the body landmark and ball detection overlays don’t show up when the video is rewinded by a few seconds or the entire video is restarted.
- Many parts of the application work better for a clear camera view that shows the complete court and all locations the player could potentially move to. This can only be achieved with a mounted camera that remains perfectly still. When the video is recorded by hand, there are many slight movements that may shift pixels by 1 or 2, but this renders the motion detection for the ball completely useless. Even with a fixed camera, factors like wind or an unstable fence can shift the camera.

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**— 6/20/26 —**

Achievements

- Algorithm modifications:
  - Ball detection
    - Initially used HSV color thresholds (yellow/green), which was too strict and didn't detect the ball.
    - Switched to RGB thresholds:
      - g > 180 && g > r + 20 && g > b + 20 && r > 120 (still too strict)
      - g > 160 && g > r + 10 && g > b + 10 && r > 80 (false positives on green tennis court surface)
      - g > 170 && g > r + 12 && g > b + 12 && r > 100 (started triggering on edges of white court lines)
    - Switched to detecting through motion, scanning for groups of pixels that change color rapidly in consecutive frames. Large areas of motion are ignored, and the area surrounding the body landmarks (player) is ignored.
    - Motion detection was very unreliable and confused the opponent's movement with the ball. I tried to block out the opponent's movement by ignoring areas that experience constant motion lasting past the amount of time a ball would take to travel through the area, but the opponent is constantly moving, so still detected.
    - The ball has a maximum velocity and must travel in a continuous path. I attempted to simplify the detector by only checking the region surrounding the ball's last detected position for new motion. However, false positives cause this region to expand and the change becomes ineffective.
    - Switched to a combination of motion and color. First detect areas that experience two rapid color changes (original -> ball -> original again), with the middle color falling in the yellow/green range. Continue to ignore large groups of pixels and set a maximum velocity on the ball movement, preventing random jumps from one point on the screen to another.
    - When the ball travels through shadow or over the blue/green court, its color changes from bright yellow to light blue/green or even a dark grey. I dropped the tennis ball color check and just checked each pixel for two consecutive color changes, with the initial and final color being similar.
    - Going back to continuity, a pixel that experiences motion can only be considered as a possible ball if there was motion in the surrounding area in the previous few frames. This slightly reduced the number of false positives that appeared on the edges of the video.
- Feature improvements:
  - Initially toast notifications appeared above the video whenever a ball was detected, but quickly stacked up and blocked the video. I changed the indicator to a constant label on the bottom, next to the playback speed dropdown, that turns green when a ball is detected.

    - To help with debugging the newly added ball detector, every 3 frames a blue circle is drawn over the video wherever a tennis ball is detected, tracing the ball's path. An HUD display showed the RGB values of any pixel that is hovered over, as well as the number of pixels that experience motion (color change).
    - Whenever a new video file is opened, the feedback side panel starts out blank until the user watches the first stroke. Previously, it would show feedback on a stroke from the scan before the user even watched the video.

Issues

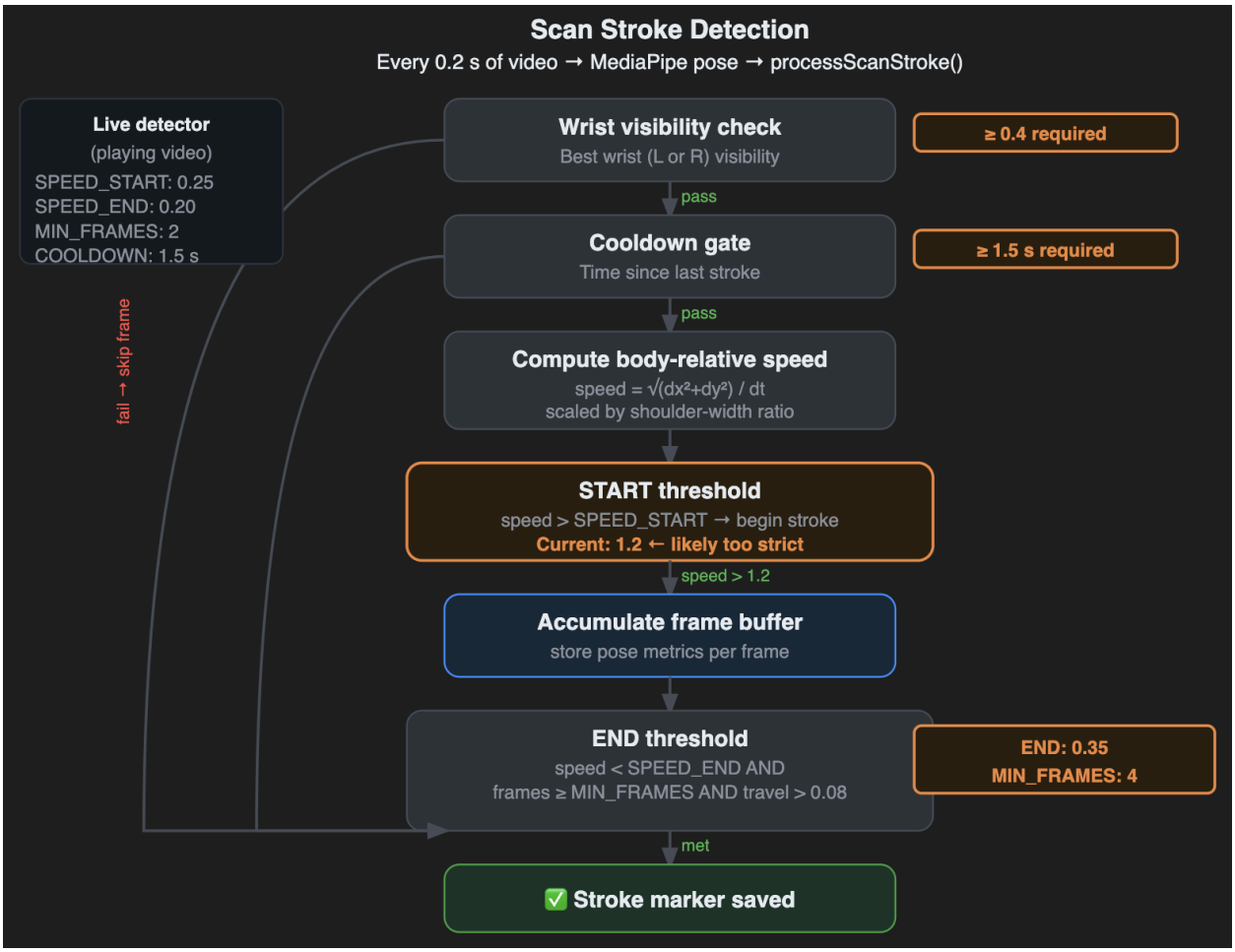
- Whenever the video file is switched while the video is playing, the play/pause button and playback speed don't reset. The button shows unpaused, and the speed dropdown shows the most recently selected speed even though the video plays at 1x speed.

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**— 6/16/26 —**

Achievements

- Algorithm modifications:
    - Stroke detection
        - Changes applied:
            - Start speed: 1.2 -> 0.65
            - End speed: 0.35 -> 0.25
            - Min number of frames: 4 -> 3
            - Required wrist visibility: 0.4 -> 0.3
            - Swing travel distance: 0.08 -> 0.05
- Feature improvements:
    - Sometimes the body landmark detector mistakenly places dots, causing unrealistic angle measurements that result in extremely low stroke scores. To avoid this, I defined plausible bounds for each angle measurement. When the measured angle exceeds this, there will be a warning ("Possibly inaccurate") and the measurement will have a decreased weight in the score calculation.
        - Elbow: 15°–180°
        - Knee: 70°–185°
        - Shoulder tilt: 0°–60°
        - Trunk lean: 0°–45°
    - The plausible bounds can help in a lot of cases, but when the player's body is completely incorrectly detected (for example, shoulders are above head) the score will be inaccurate. I added a warning (similar to the out of frame warning) that tells the user detection and analysis may be inaccurate when basic anatomical ordering (nose, shoulders, hips, knees, ankles) goes out of order. If this warning triggers during a stroke, the stroke's score will have a small indicator.
    - I incorporated a tennis ball detector that can be helpful in identifying strokes (when the ball changes direction) and body orientation. However, with unclear




files the ball is sometimes hard to detect, and the algorithm needs to be adjusted for serves, when the direction change is less extreme.

Issues

- Feedback updates can be triggered by both the detector (when speed drops) and when the stroke ends (end time + 0.2), sometimes causing the feedback panel to update multiple times while a single stroke is being hit.
- When the camera angle is from the side, and the player's shoulders are square, the detector can't tell if the player is facing towards or away from the camera. This makes it unable to differentiate forehands from backhands.

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**— 6/11/26 —**

Achievements

- Algorithm modifications:
    - Stroke end time calculation
        - Initial algorithm: stroke start time + 1 s (assumes constant swing length for all types of strokes)
        - Improved: stroke end time + 0.2 s

    - Continue working on stroke classification
        - In a serve, the wrist speed drops as the player reaches trophy position, causing the detector to end the stroke. The wrist never gets above the shoulder and the stroke is never classified as a serve/overhead. I added a pending serve check watch to see if the wrist goes above shoulder in the next 2.5 seconds after a stroke ends.
        - When the player's body is turned to the side, so that the shoulders are perpendicular to the camera, a large racquet takeback can cross to the opposite side of the body, causing misclassification between forehands and backhands. Changed to find the wrist's position at its maximum horizontal distance from the shoulder midpoint and see which side of the body it's on.
- Feature improvements:
    - A Rescan option, for when the background scan has already been conducted, to implement new algorithm updates.
    - Undo button at the bottom of the score panel in case a stroke is mistakenly deleted.
    - When the player's body leaves the frame or is too far away, and the detector can't find the body landmarks, I added a warning that tells the user that no player is detected and pauses the video. There is an option to continue watching the video or skip ahead to when the player returns to the frame. This warning initially covered the entire score panel, I later changed it to be a popup bar at the bottom of the panel that doesn't pause the video. The player can choose to dismiss the warning or skip ahead to the next visible time.
    - A right/left-handedness switch that flips forehand/backhand classification for left-handed players.
- I've been using GoPro footage of a tennis match to test the analyzer, but the camera angle and video quality are not high. Sometimes the player moves out of frame, or far away from the camera, making it difficult to detect body landmarks. In order to test and finetune the analyzer more easily, I filmed closeup videos of myself performing simple strokes with good and bad technique.

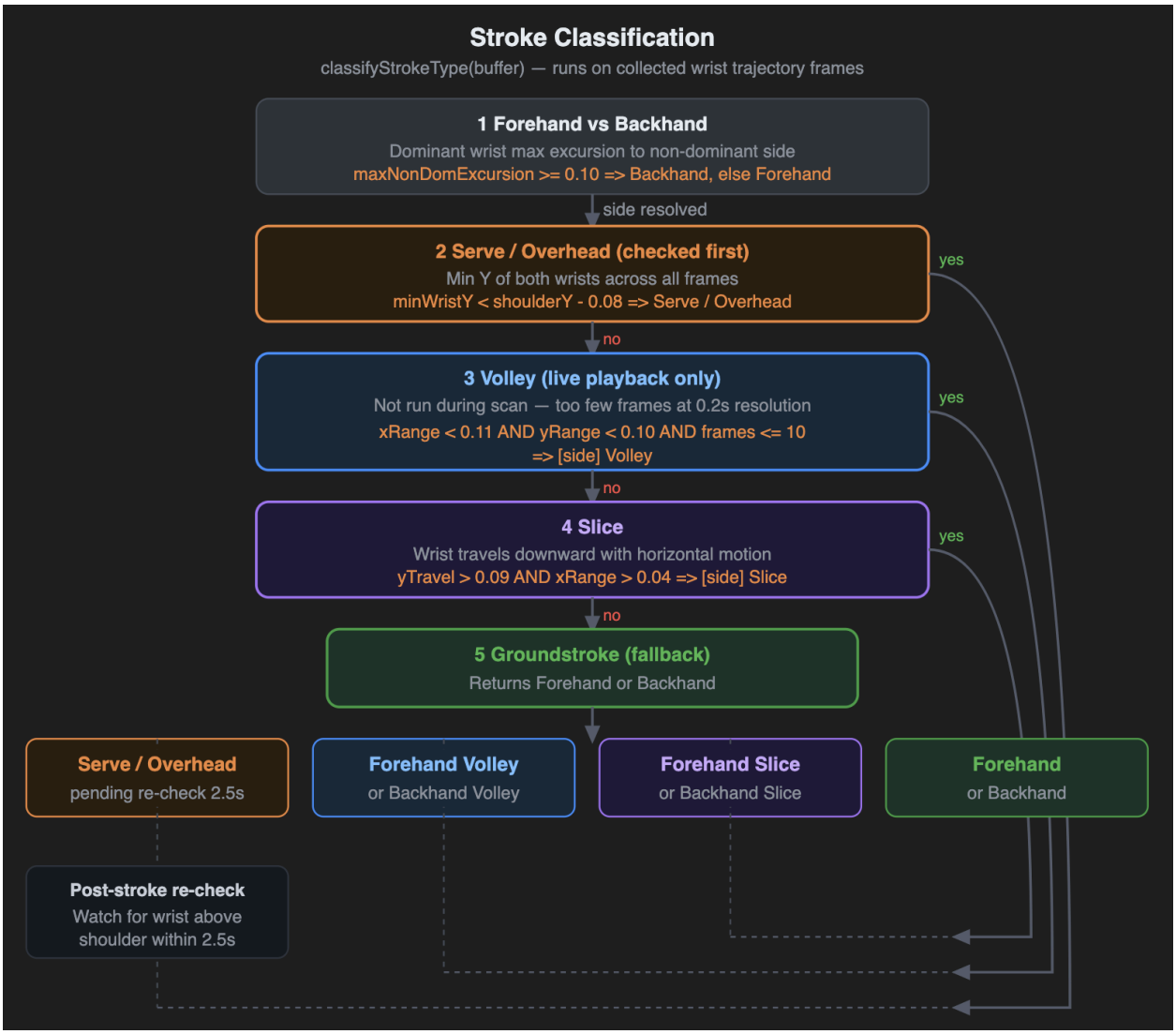
Issues

- When the initial background scan is skipped, there is a bug that causes the Auto-pause feature to not work even when the user has selected it.
- When the dominant wrist becomes less visible (like when blocked by the player's body), the detector switches to using the non-dominant wrist, resulting in inaccurate analysis.
- The detector "starts" a stroke when the wrist speed reaches a threshold, but this often excludes the slower preparation/takeback in a swing. This was improved by maintaining a 2 second wrist history. When the wrist speed increases, the history is searched for the time when the arm was at neutral position (wrist aligned with shoulder) and uses that as the start time.
- Similarly, the stroke ends early when the follow-through is slow. To fix this, the stroke only ends when the wrist has traveled a certain distance after the wrist speed drops below the stroke threshold.

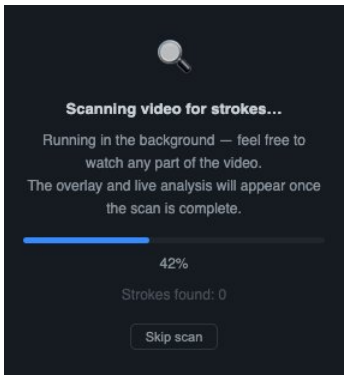
**— 6/4/2026 —**

Achievements

- Algorithm modifications:
    - Try to make stroke classification more accurate
        - Initial algorithm:
            - compact swing (limited x and y travel) -> volley
            - wrist above shoulder and large range -> serve/overhead
            - downward travel -> slice
            - none of the above -> groundstroke
        - The scan only looks at a few frames of the serve, detecting a small range, which classifies as volley.



        - Forehand/backhand distinction was incorrectly comparing the wrist position to hips, causing almost all strokes to classify as forehand.




- Feature improvements:
    - I moved the scan to the background so the user doesn't have to wait for the scan to complete. There is a scan progress bar and a stroke counter. The user can freely use the playback tools to watch the video.
    - I stored the stroke markers (on the progress bar) in Firestore so the user doesn't have to rescan after opening the website again.
    - I removed the angle numbers from the video screen because they overlapped and were unreadable. The only thing that appears on the video screen are dots, for body landmarks, and connecting lines for limbs.
    - I added an Auto-pause feature, which if selected pauses the video whenever a stroke finishes so the user can read the analysis and feedback. Once done, the user can click "Continue" and the video unpauses until the next stroke is finished. If the analyzer incorrectly detects a stroke, the user can select "Not a stroke" and the marker disappears.

Issues

- The stroke counter remained at 0 throughout the entire scan. This was fixed after adjusting the stroke detection requirements (wrist speed threshold).
- The timing of the pause had to be adjusted because the detector tends to stop the stroke right after the ball is hit, before the follow-through is over.
- The background scan adds markers to the progress bar, but watching the video after the scan continues to add new markers. This is due to different frames being sampled, causing different detected stroke start times and scores. I changed the background scan to only detect strokes, adding grey markers, while the real-time playback assigned scores to the markers. This was unreliable because new strokes were still being detected. I switched back to having the background scan detect strokes and assign scores, storing these in the scan buffer. The real-time playback takes scores and feedback directly from the buffer.
- Sometimes the stroke analysis panel doesn't update quickly enough, and it would skip a few strokes if they were close together. Fixed to update whenever a new stroke starts.

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**— 5/31/2026 —**

Achievements (more advanced features)

- While the analyzer scans through the video, small tick marks appear on the video progress bar to show detected strokes. The marks are colored green, yellow, or red

according to the stroke's score. The user can click on a mark to watch the associated stroke.
- A background scan plays the video at 4x speed and samples frames at regular intervals, so the user doesn't have to watch the entire video for the analyzer to finish. There is also a Skip Scan option that allows the user to watch the video and see feedback appear in real time.
- A stroke classification feature categorizes each stroke as forehand or backhand groundstroke/volley/slice or serve/overhead.

Issues

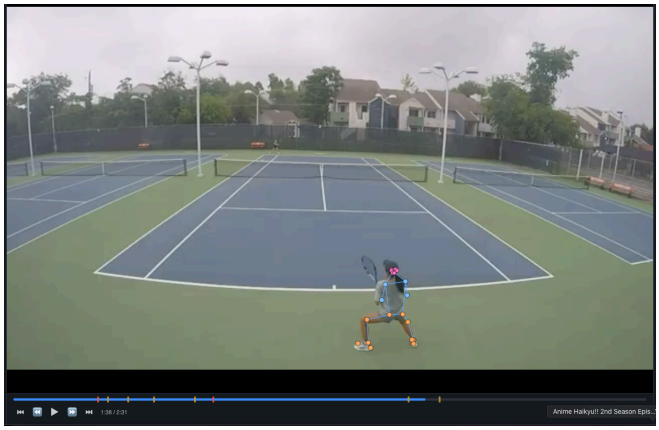
- Detection is still inaccurate so the marks don't line up with actual strokes, there are much more marks than strokes.
- Almost every stroke is being classified as a forehand volley.

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**— 5/28/26 —**

Achievements (initial framework setup)

- I set up a simple website with initial features including video file upload and basic playback functions. Strokes are detected, analyzed, and scored, with feedback appearing in a side panel.
    - **Firebase** authentication allows users to log in and store uploaded files. A detailed instruction page to set up Firebase appears for first-time users.
    - **MediaPipe Pose** is used for body detection and to draw a skeleton over the player's body in the video.
        - MediaPipe Pose is a machine learning model from Google that detects a person's body in a video frame and outputs 33 landmarks (shoulders, elbows, wrists, hips, knees, ankles, etc.). Each landmark has x/y coordinates and a visibility score. They are used to calculate wrist speed (normalized using shoulder width) and joint angles (elbow angle, knee bend, shoulder tilt, and trunk lean).
    - **COCO-SSD** is used to detect the racquet. This model is run every 20 frames and draws a bounding box around the racquet when it detects it with confidence > 60%.
        - COCO-SSD (Common Objects in Context Single Shot MultiBox Detector) is an object detection model trained on 80 everyday object categories (including "tennis racket").

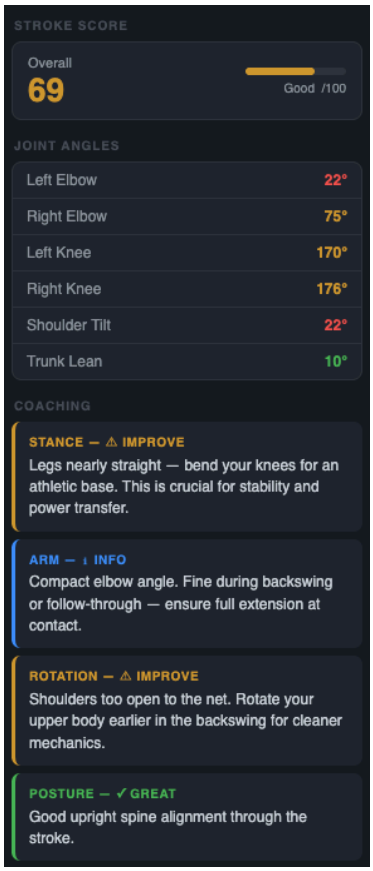


- I chose an initial set of measurements, taken from the

Mediapipe Pose landmarks, that are used to score the stroke. Each metric contributes a number of points toward 100 based on how close the measured angle is to the ideal range (inside: 100, +- 15%: 55, outside: 12).

- Elbow angle: taken from the right arm, ideally from 95 to 158 degrees, weight of 30.
- Knee bend: averaged from the right and left legs, should be 115 to 155 degrees, weight of 30
- Shoulder tilt: the angle the shoulder line makes with the horizontal, should be from 0 to 10 degrees, weight of 20.
- Trunk lean: the angle between the line joining the shoulder midpoint to the hip midpoint and the vertical, ideally 0 to 13 degrees, weight of 20.

- I wrote a default set of feedback that is connected to the measurements.
  - Arm (elbow)
    - 162°: warn — arm too straight
    - 100°–162°: good — natural bend
    - 65°–100°: info — compact, fine during backswing
    - < 65°: warn — too bent, not extending through ball
  - Stance (knee bend)
    - 168°: warn — legs nearly straight
    - 152°–168°: warn — bend more
    - 115°–152°: good — solid athletic position
    - < 115°: info — very deep, check mobility
  - Rotation (shoulder tilt)
    - < 7°: good — well rotated
    - 7°–16°: info — moderate rotation
    - 16°: warn — shoulders too open
  - Posture (trunk lean)
    - ≤ 11°: good — upright alignment
    - 11°–23°: info — slight lean, monitor balance
    - 23°: warn — significant lean




Issues

- The body landmark detection is unstable, causing the angles to be inaccurate.
- Racquet detection is unreliable. This causes stroke detection to be wrong, with a single stroke sometimes having multiple scores.
- The measurements have many limitations. For example, the angles are averaged throughout the entire swing, which includes preparation and follow-through, making the feedback less accurate. The feedback also doesn't distinguish between different types of strokes, which may require different mechanics and ranges.